Project: Worked examples
Title: Voltage drop and I²R loss
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1. Scope and method

Voltage drop along a circuit in volts and per cent of the nominal voltage, the conductor power loss, and an estimate of the conductor operating temperature, for DC, single-phase and three-phase AC, from resistivity and cross-section (with temperature correction), from tabulated R and X per kilometre, or from a mV/A/m figure; parallel conductors per phase; the drop limit is a user input (the regulatory limits live in BS 7671 Appendix 4 and Regulation 525 and are not embedded).

WHAT IT CALCULATES	RELATION
Conductor resistance at its operating temperature, for n conductors in parallel	$R = \rho_T L/(A\,n)$, with $\rho_T = \rho_{20}\,[1 + \alpha\,(T_c - 20)]$
Resistance and reactance from per-length figures instead	$R = R' L/n$; $X = X' L/n$
Voltage drop on a DC circuit	$\Delta V = 2\,I\,R$
Voltage drop on a single-phase AC circuit	$\Delta V = 2\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop on a three-phase AC circuit	$\Delta V = \sqrt{3}\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop from a tabulated mV/A/m figure, which already carries the 2 or √3	$\Delta V = (\text{mV/A/m})\,I\,L / 1000$
Power lost in the cable	$P_{loss} = 2\,I^2\,R$ (DC and single-phase); $3\,I^2\,R$ (three-phase)
Estimated conductor temperature at this current	$T_{c,est} = T_{amb} + (T_{rated} - T_{amb})(I/I_z)^2$

Basis: Voltage drop in consumers' installations[1](525); Voltage drop (mV/A/m tables and the temperature-correction method)[1](Appendix 4, section 6); Resistance, resistivity and temperature coefficient[2](Ch. on DC and AC circuits).

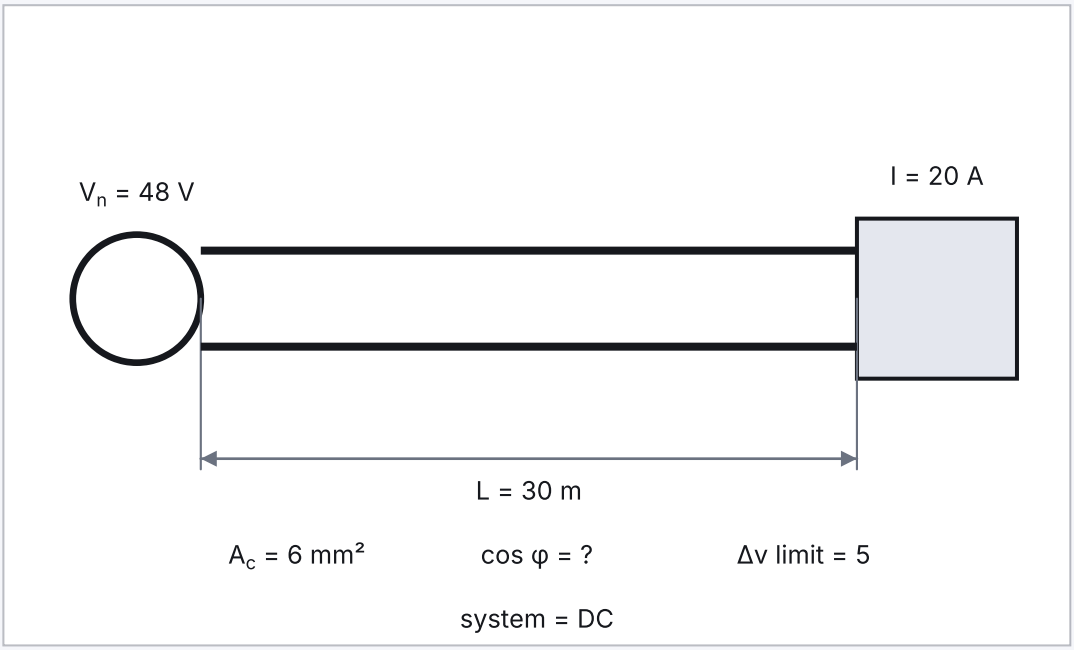


Figure 1 — A DC circuit: the supply, the two conductors of route length L, and the load current I. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
system	System	DC		assumption: Design convention.	any
V_n	Nominal voltage (line-to-line for three-phase)	48	V	assumption: Design convention.	12 … 1000 V
I	Design current	20	A	spec: Load schedule DC-02: 20 A	0.1 … 5000 A
L	Route length (one way)	30	m	drawing: DRG-E-301 rev A	0.1 … 5000 m
$\cos_{phi}$	Power factor (AC)	0.85		default	0.5 … 1
n_{par}	Parallel conductors per phase	1		assumption: Design convention.	1 … 6
data	Conductor data from	resistivity and area		assumption: Design convention.	any
ρ_{20}	Resistivity at 20 °C, the effective value for the conductor resistance (default 0.0194 Ω·mm²/m: the largest class 2 maximum conductor resistance of the cable standard for 1.5 to 120 mm², so the drop is not below the cable tables; pure annealed copper is 0.01724, aluminium 0.0282 Ω·mm²/m)	0.01724	Ω·mm²/m	reference: Reference record: annealed copper (Hughes)	0.01 … 0.05 Ω·mm²/m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
α	Temperature coefficient of resistance	0.00393		reference: Reference record: annealed copper (Hughes)	0.003 … 0.0045
A_c	Conductor cross-section	6	mm²	spec: Cable schedule CS-02: 6 mm² Cu	1 … 1000 mm²
T_c	Conductor operating temperature for the resistance	70	°C	assumption: Design convention.	20 … 90 °C
X_{km}	Reactance per km per conductor	0	Ω/km	assumption: Design convention.	0 … 1 Ω/km
dv_{limit}	Voltage-drop limit	5		spec: Battery system spec BS-1: 5 % at the load	1 … 10
T_{rated}	Conductor rated temperature	70	°C	default	60 … 90 °C
T_{amb}	Ambient temperature	30	°C	default	0 … 60 °C

3. Assumptions

A1. Balanced load; conductor reactance from the stated X′ (zero when using resistivity alone unless X_{km} is given); the drop is the arithmetic (R cos φ + X sin φ) form. (Standard engineering approximation.)
A2. A mV/A/m figure is used as tabulated, with its own temperature and system convention. (Cable table convention.)
A3. The conductor temperature estimate scales the rated rise with $(I/I_z)^2$; I_z is the user's tabulated value for the installation method and corrections. When the estimate is above T_c the resistance and the drop are computed at the estimate. (BS 7671 Appendix 4 approach, formula only.)
A4. system (System) is taken as DC. (Design convention.)
A5. V_n (Nominal voltage (line-to-line for three-phase)) is taken as 48 V. (Design convention.)
A6. n_{par} (Parallel conductors per phase) is taken as 1. (Design convention.)
A7. data (Conductor data from) is taken as resistivity and area. (Design convention.)
A8. T_c (Conductor operating temperature for the resistance) is taken as 70 °C. (Design convention.)
A9. X_{km} (Reactance per km per conductor) is taken as 0 Ω/km. (Design convention.)

4. Working

[9] $\sin_{phi} = 0$
$= 0$
sin φ
[15] $\rho_T = \rho_{20} * (1 + \alpha * (T_c - 20\ \text{degC}) / (1\ \text{K}))$
$= 0.01724\ \text{ohm·mm²/m} \times (1 + 0.00393 \times (70\ \text{degC} - 20\ \text{degC}) / 1\ \text{K})$
$= 0.020628\ \text{ohm·mm²/m}$
Resistivity at the operating temperature
[16] $R = \rho_T * L / (A_c * n_{par})$
$= 0.020628\ \text{ohm·mm²/m} \times 30\ \text{m} / (6\ \text{mm²} \times 1)$
$= 0.10314\ \text{ohm}$
Conductor resistance (one conductor, parallel paths combined)
[17] $X = X_{km} * L / n_{par}$
$= 0\ \text{ohm/km} \times 30\ \text{m} / 1$
$= 0\ \text{ohm}$
Conductor reactance

[18] dV = 2 * I * R
= 2 × 20 A × 0.10314 ohm
= 4.1255 V
Voltage drop
[19] dV_{pct} = 100 * dV / V_n
= 100 × 4.1255 V / 48 V
= 8.5949
Voltage drop, per cent
[20] P_{loss} = 2 * I^2 * R
= 2 × (20 A)^2 × 0.10314 ohm
= 82.511 W
Conductor power loss
[21] u_{dv} = dV_{pct} / dv_{limit}
= 8.5949 / 5
= 1.719
Drop utilisation against the limit

5. Results

SYMBOL	VALUE	DESCRIPTION
R	0.1031 Ω	Resistance
X	0 Ω	Reactance
dV	4.126 V	Voltage drop
dV_{pct}	8.5949	Voltage drop (%)
P_{loss}	82.51 W	Power loss

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Voltage drop ≤ limit	u_{dv} = 1.72	u_{dv} ≤ 1	1.72	[1](525)

7. Applicability

None.

Defaults used (inputs left empty):
cos_{phi}: not entered, so the default of 0.85 is used.
T_{rated}: not entered, so the default of 70 °C is used.
T_{amb}: not entered, so the default of 30 °C is used.

8. Verification

Template elec.voltage-drop@1.3.0 Content hash 1dc7d4b34be3c142 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.